04/21/26

1a. Output for RunnableExample.java:

```
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.
exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964
ada4\redhat.java\jdt_ws\Week 14_b61c3775\bin' 'RunnableExample'
Sum (Runnable): 5050
```

1b. Output for ThreadExample.java:

```
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-
hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\vgmat\AppData\Roaming\Code\User\workspaceSt
orage\7f86c77f543cf337a049b1148964ada4\redhat.java\jdt_ws\Week 14_b61c3775\bin' 'ThreadExample'
Sum (Thread): 5050
```

2a. Output for CounterVariants.java (Will always be 10000):

```
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.
exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964
ada4\redhat.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants'
Mode: 1
Final Counter: 10000
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> ^C
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  c:; cd 'c:\Users\vgmat\Downloads\Principles of Programming Concepts
\Week 14'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users
\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964ada4\redhat.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants
'
Mode: 1
Final Counter: 10000
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> ^C
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  c:; cd 'c:\Users\vgmat\Downloads\Principles of Programming Concepts
\Week 14'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users
\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964ada4\redhat.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants
'
Mode: 1
Final Counter: 10000
```

2b. Same as 2a.

2c. If the main thread runs without joining any thread, the output will be less than 10000, because the main thread prints too early. It can be 10000 sometimes.

```
Mode: 2
Final Counter: 50
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> ^C
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  c:; cd 'c:\Users\vgmat\Downloads\Principles of Pro
gramming Concepts\Week 14'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsIn
ExceptionMessages' '-cp' 'C:\Users\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964ada4\redhat
.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants'
Mode: 2
Final Counter: 83
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> ^C
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  c:; cd 'c:\Users\vgmat\Downloads\Principles of Pro
gramming Concepts\Week 14'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsIn
ExceptionMessages' '-cp' 'C:\Users\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964ada4\redhat
.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants'
Mode: 2
Final Counter: 87
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> ^C
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  c:; cd 'c:\Users\vgmat\Downloads\Principles of Pro
gramming Concepts\Week 14'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsIn
ExceptionMessages' '-cp' 'C:\Users\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964ada4\redhat
.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants'
Mode: 2
Final Counter: 91
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> ^C
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  c:; cd 'c:\Users\vgmat\Downloads\Principles of Pro
gramming Concepts\Week 14'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsIn
ExceptionMessages' '-cp' 'C:\Users\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964ada4\redhat
.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants'
Mode: 2
Final Counter: 69
```

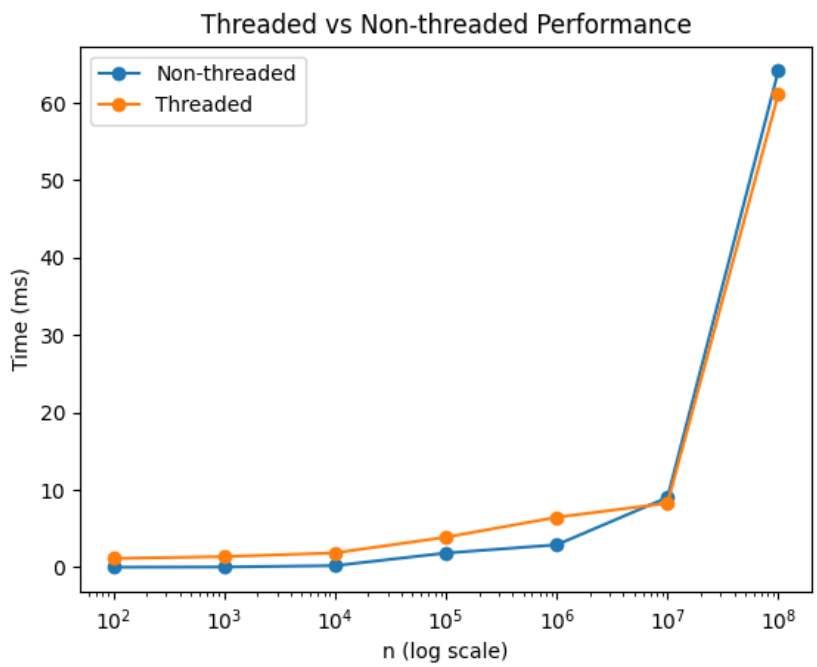
2d. If the increment method is not synchronized, it will never reach 10000.

```
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> ^C
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  c:; cd 'c:\Users\vgmat\Downloads\Principles of Programming Concepts
\Week 14'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users
\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964ada4\redhat.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants
'
Mode: 3
Final Counter: 9566
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> ^C
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  c:; cd 'c:\Users\vgmat\Downloads\Principles of Programming Concepts
\Week 14'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users
\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964ada4\redhat.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants
'
Mode: 3
Final Counter: 9790
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> ^C
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  c:; cd 'c:\Users\vgmat\Downloads\Principles of Programming Concepts
\Week 14'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users
\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964ada4\redhat.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants
'
Mode: 3
Final Counter: 9777
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> ^C
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14>  c:; cd 'c:\Users\vgmat\Downloads\Principles of Programming Concepts
\Week 14'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.4.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users
\vgmat\AppData\Roaming\Code\User\workspaceStorage\7f86c77f543cf337a049b1148964ada4\redhat.java\jdt_ws\Week 14_b61c3775\bin' 'CounterVariants
'
Mode: 3
Final Counter: 9659
```

3. Output for SumSquares.java:

```
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> java SumSquares 100
n = 100
Non-threaded time: 2200
Threaded time: 1135400
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> java SumSquares 1000
n = 1000
Non-threaded time: 28000
Threaded time: 1368300
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> java SumSquares 10000
n = 10000
Non-threaded time: 204400
Threaded time: 1848900
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> java SumSquares 100000
n = 100000
Non-threaded time: 1829400
Threaded time: 3875700
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> java SumSquares 1000000
n = 1000000
Non-threaded time: 2893300
Threaded time: 6454700
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> java SumSquares 10000000
n = 10000000
Non-threaded time: 9011600
Threaded time: 8274300
PS C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 14> java SumSquares 100000000
n = 100000000
Non-threaded time: 64144500
Threaded time: 61127200
```

1	n	Non-th	Thread	Non-th	Thread
2	100	2200	1135400	0.0022	1.1354
3	1000	28000	1368300	0.028	1.3683
4	10000	204400	1848900	0.2044	1.8489
5	100000	1829400	3875700	1.8294	3.8757
6	1000000	2893300	6454700	2.8933	6.4547
7	10000000	9011600	8274300	9.0116	8.2743
8	100000000	64144500	61127200	64.1445	61.1272



The results show that for small values of n, the non-threaded implementation performs significantly faster than the threaded version. This is because threads introduce overhead that's unnecessary in smaller values. As n increases, the gap between the two approaches begins to close, and for very large values, the threaded version becomes slightly faster. This occurs because the workload is large enough to benefit from parallel processing.